

8.3. Quantum Electrodynamics (QED)

8.3.1. Local Gauge Invariance

We use a metric with signature +2 in this section.

The Dirac equation for an electron is:

$$\mathcal{L} = \bar{\psi}(i\hbar c\gamma^\mu\partial_\mu - mc^2)\psi$$

Physical quantities like a charge density ($\bar{\psi}\psi$) or a current ($\bar{\psi}\gamma^\mu\psi$), are all invariant if we add a local phase $\Lambda(x)$ to the field (this is called a local U(1) gauge transformation):

$$\begin{aligned}\psi(x) &\rightarrow e^{iq\Lambda(x)/\hbar}\psi(x) \\ \bar{\psi}(x) &\rightarrow \bar{\psi}(x)e^{-iq\Lambda(x)/\hbar}\end{aligned}$$

Where q is a parameter that measures the strength of the phase transformation (this will be later interpreted as a charge, for example for electrons $q = -|e|$) and $\hbar$ is the Planck constant. And so we require that the Lagrangian is also invariant under the local gauge transformation, because there is no experiment that would change if this local gauge transformation is applied on the wave functions. By putting this gauge transformation into the Lagrangian density, we obtain:

$$\begin{aligned}\mathcal{L} &\rightarrow \bar{\psi}e^{-iq\Lambda(x)/\hbar}(i\hbar c\gamma^\mu\partial_\mu - mc^2)e^{iq\Lambda(x)/\hbar}\psi = \\ &= \bar{\psi}(i\hbar c\gamma^\mu(\partial_\mu + iq\partial_\mu\Lambda(x)/\hbar) - mc^2)\psi\end{aligned}$$

The reason the Lagrangian is not invariant is due to the derivative, which does not transform covariantly under a local gauge transformation:

$$\begin{aligned}\bar{\psi}\partial_\mu\psi &\rightarrow \bar{\psi}e^{-iq\Lambda(x)/\hbar}\partial_\mu e^{iq\Lambda(x)/\hbar}\psi = \\ &= \bar{\psi}(\partial_\mu + iq\partial_\mu\Lambda(x)/\hbar)\psi \neq \bar{\psi}\partial_\mu\psi\end{aligned}$$

In order to make the derivative transform covariantly (and thus the Lagrangian gauge invariant), we have to introduce a gauge field, in this case a vector field $A_\mu(x)$, as follows:

$$D_\mu = \partial_\mu - \frac{i}{\hbar}qA_\mu \tag{8.3.1.1}$$

and the field A_μ must transform as $A_\mu \rightarrow A_\mu + \partial_\mu\Lambda(x)$. At this level, we are free to choose either plus or minus sign in (8.3.1.1), since the sign change can be absorbed in the definition of the A_μ field without loss of generality (if we change the sign, the field transformation then changes to $A_\mu \rightarrow A_\mu - \partial_\mu\Lambda(x)$). In the +2 metric signature we chose a minus sign, so that A_μ coincides with the usual definition of the electromagnetic 4-potential:

$$\begin{aligned}
D_\mu &= \partial_\mu - \frac{i}{\hbar} q A_\mu \\
-i\hbar D_\mu &= -i\hbar \partial_\mu - q A_\mu \\
m\hat{v}_\mu &= \hat{p}_\mu - q A_\mu \\
m\hat{\mathbf{v}} &= \hat{\mathbf{p}} - q\mathbf{A}
\end{aligned}$$

With signature -2, we must choose a plus sign and the identification goes as follows:

$$\begin{aligned}
D_\mu &= \partial_\mu + \frac{i}{\hbar} q A_\mu \\
i\hbar D_\mu &= i\hbar \partial_\mu - q A_\mu \\
m\hat{v}_\mu &= \hat{p}_\mu - q A_\mu \\
m\hat{\mathbf{v}} &= \hat{\mathbf{p}} - q\mathbf{A}
\end{aligned}$$

And we obtain the same final equation. So the kinematic momentum is equal to canonical momentum minus charge times the gauge field. The last expression is independent of a metric signature, and that is what is e.g. in the kinetic term of a Schrödinger or Pauli equation (with the minus sign in $\hat{\mathbf{p}} - q\mathbf{A}$). We derive the non-relativistic limit rigorously later, but it gives the same result. At this level we just have to make sure we choose the correct sign in (8.3.1.1), depending on the metric signature, otherwise we would get the electromagnetic 4-potential with the opposite sign (the sign of A_μ is ultimately just a convention, but later we want to get the same equations as everybody else).

Another unrelated convention is in choosing the sign of the parameter q . We have chosen it to coincide with an electric charge (negative for electrons). Some authors choose q to be positive for electrons, then one must flip the sign in (8.3.1.1).

We will continue using the +2 signature in the rest of the section.

The operator $D_\mu = \partial_\mu - \frac{i}{\hbar} q A_\mu$ is called a covariant derivative, because it does not change a form (is invariant) under a local gauge transformation:

$$\begin{aligned}
\bar{\psi} D_\mu \psi &= \bar{\psi} (\partial_\mu - \frac{i}{\hbar} q A_\mu) \psi \\
\rightarrow \bar{\psi} e^{-iq\Lambda(x)/\hbar} (\partial_\mu - \frac{i}{\hbar} q (A_\mu + \partial_\mu \Lambda(x))) e^{iq\Lambda(x)/\hbar} \psi &= \\
= \bar{\psi} (\partial_\mu - \frac{i}{\hbar} q A_\mu - \frac{i}{\hbar} q \partial_\mu \Lambda(x) + iq \partial_\mu \Lambda(x)/\hbar) \psi &= \\
= \bar{\psi} (\partial_\mu - \frac{i}{\hbar} q A_\mu) \psi &= \bar{\psi} D_\mu \psi
\end{aligned}$$

Then the Lagrangian

$$\mathcal{L} = \bar{\psi} (i\hbar c \gamma^\mu (\partial_\mu - iq A_\mu/\hbar) - mc^2) \psi \quad (8.3.1.2)$$

is also gauge invariant:

$$\begin{aligned}
\mathcal{L} &\rightarrow \bar{\psi} e^{-iq\Lambda(x)/\hbar} (i\hbar c \gamma^\mu (\partial_\mu - iq A_\mu/\hbar - iq \partial_\mu \Lambda(x)/\hbar) - mc^2) e^{iq\Lambda(x)/\hbar} \psi = \\
&= \bar{\psi} (i\hbar c \gamma^\mu (\partial_\mu - iq A_\mu/\hbar - iq \partial_\mu \Lambda(x)/\hbar + iq \partial_\mu \Lambda(x)/\hbar) - mc^2) \psi = \\
&= \bar{\psi} (i\hbar c \gamma^\mu (\partial_\mu - iq A_\mu/\hbar) - mc^2) \psi
\end{aligned}$$

The Lagrangian (8.3.1.2) can also be written as:

$$\begin{aligned}\mathcal{L} &= \bar{\psi}(i\hbar c\gamma^\mu(\partial_\mu - \frac{i}{\hbar}qA_\mu) - mc^2)\psi = \\ &= \bar{\psi}(i\hbar c\gamma^\mu\partial_\mu - mc^2)\psi + qc\bar{\psi}\gamma^\mu\psi A_\mu\end{aligned}$$

We can see that the condition of a local gauge invariance requires an interaction with a vector field A_μ . Now we need to add the kinetic term for the field A_μ :

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

The mass term $\frac{1}{2}m^2A_\mu A^\mu$ is not gauge invariant, and so we have to set $m = 0$. Here is the full Lagrangian:

$$\mathcal{L} = \bar{\psi}(i\hbar c\gamma^\mu\partial_\mu - mc^2)\psi + qc\bar{\psi}\gamma^\mu\psi A_\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

This is a Lagrangian for an electron and a massless vector boson (photon) of spin 1. We can introduce a current $j^\mu = c\bar{\psi}\gamma^\mu\psi$, then the Lagrangian density becomes:

$$\mathcal{L} = \bar{\psi}(i\hbar c\gamma^\mu\partial_\mu - mc^2)\psi + qj^\mu A_\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

For an electron, we can set $q = -e$, where e is the elementary charge (e is positive).

8.3.2. QED Lagrangian

We use a metric with signature -2 in this section.

The QED Lagrangian density is

$$\mathcal{L} = \bar{\psi}(i\hbar c\gamma^\mu D_\mu - mc^2)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

where

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

and we must choose a plus sign in (8.3.1.1) since we use the -2 signature:

$$D_\mu = \partial_\mu + \frac{i}{\hbar}eA_\mu$$

e is the charge (negative for electrons $e = -|e|$).

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

is the electromagnetic field tensor. It's astonishing, that this simple Lagrangian can account for all phenomena from macroscopic scales down to something like 10^{-13} cm. So it's not a surprise that Feynman, Schwinger and Tomonaga received the 1965 Nobel Prize in Physics for such a fantastic achievement.

Plugging this Lagrangian into the Euler-Lagrange equation of motion for a field, we get:

$$(i\hbar c\gamma^\mu D_\mu - mc^2)\psi = 0$$

$$\partial_\nu F^{\nu\mu} = -ec\bar{\psi}\gamma^\mu\psi$$

The first equation is the Dirac equation in the electromagnetic field and the second equation is a set of Maxwell equations ($\partial_\nu F^{\nu\mu} = -ej^\mu$) with a source $j^\mu = c\bar{\psi}\gamma^\mu\psi$, which is a 4-current coming from the Dirac equation.

8.3.3. Magnetic moment of an electron

In this section we derive the order- α correction to the magnetic moment of an electron.

We start by computing the electron vertex function for the process $\gamma(q) \rightarrow e^+(p) + e^-(p')$:

$$iM = ie^2 (\bar{u}(p')\Gamma^\mu(p', p)u(p)) \frac{1}{q^2} (\bar{u}(k')\gamma_\mu u(k))$$

where k corresponds to some heavy target. If A_μ^{cl} is a fixed classical potential, we get:

$$iM2\pi\delta(p^{0'} - p^0) = -ie\bar{u}(p')\Gamma^\mu(p', p)u(p)A_\mu^{\text{cl}}$$

Using general arguments (Lorentz invariance, parity-conservation, Ward identity) we can always write Γ^μ as:

$$\Gamma^\mu(p', p) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m} F_2(q^2)$$

where F_1 and F_2 are unknown functions of $q^2 = (p' - p)^2 = -2p' \cdot p + 2m^2$ called form factors. As we will see below, in the lowest order we get $F_1 = 1$ and $F_2 = 0$.

We can calculate the amplitude for elastic Coulomb scattering of a nonrelativistic electron from a region of nonzero electrostatic potential by setting $A_\mu^{\text{cl}}(x) = (\phi(\mathbf{x}), 0)$, then:

$$\begin{aligned} A_\mu^{\text{cl}}(q) &= (2\pi\delta(q^0)\tilde{\phi}(\mathbf{q}), 0) \\ iM2\pi\delta(p^{0'} - p^0) &= -ie\bar{u}(p')\Gamma^0(p', p)u(p)2\pi\delta(q^0)\tilde{\phi}(\mathbf{q}) \\ iM &= -ie\bar{u}(p')\Gamma^0(p', p)u(p)\tilde{\phi}(\mathbf{q}) \end{aligned}$$

If the electrostatic field is very slowly varying over a large (even macroscopic) region, $\tilde{\phi}(\mathbf{q})$ will be concentrated about $\mathbf{q} = 0$, then we can take the limit $\mathbf{q} \rightarrow 0$:

$$\begin{aligned}
iM &= -ie\bar{u}(p')\Gamma^0(p', p)u(p)\tilde{\phi}(\mathbf{q}) \\
iM &= -ie\bar{u}(p')\left(\gamma^0 F_1(q^2) + \frac{i\sigma^{0\nu}q_\nu}{2m}F_2(q^2)\right)u(p)\tilde{\phi}(\mathbf{q}) \\
iM &= -ie\bar{u}(p')\gamma^0 u(p)F_1(0)\tilde{\phi}(\mathbf{q}) \\
iM &= -ie2m\xi'^\dagger \xi F_1(0)\tilde{\phi}(\mathbf{q}) \\
iM &= -i\left(eF_1(0)\tilde{\phi}(\mathbf{q})\right)2m\xi'^\dagger \xi
\end{aligned}$$

This corresponds to the Born approximation for scattering from a potential

$$V(\mathbf{x}) = eF_1(0)\phi(\mathbf{x})$$

Thus $F_1(0)$ is the electric charge of the electron, in units of e . Since $F_1(0) = 1$ already in the first order of perturbation theory, radiative corrections to $F_1(q^2)$ must vanish at $q^2 = 0$.

Now we calculate the scattering from a static vector potential by setting $A_\mu^{\text{cl}}(x) = (0, \mathbf{A}_\mu^{\text{cl}}(\mathbf{x}))$, then:

$$\begin{aligned}
A_{\text{cl}}^\mu(q) &= (0, 2\pi\delta(q^i)\tilde{A}_{\text{cl}}^i(\mathbf{q})) \\
iM2\pi\delta(p'^i - p^i) &= ie\bar{u}(p')\Gamma^i(p', p)u(p)2\pi\delta(q^i)\tilde{A}_{\text{cl}}^i(\mathbf{q}) \\
iM &= ie\bar{u}(p')\Gamma^i(p', p)u(p)\tilde{A}_{\text{cl}}^i(\mathbf{q}) \\
iM &= ie\bar{u}(p')\left(\gamma^i F_1(q^2) + \frac{i\sigma^{i\nu}q_\nu}{2m}F_2(q^2)\right)u(p)\tilde{A}_{\text{cl}}^i(\mathbf{q})
\end{aligned}$$

In the limit $q \rightarrow 0$ this becomes:

$$\begin{aligned}
iM &= ie2m\xi'^\dagger \left(-i\epsilon^{ijk}\frac{q^j\sigma^k}{2m}(F_1(0) + F_2(0))\right)\xi\tilde{A}_{\text{cl}}^i(\mathbf{q}) \\
iM &= -ie2m\xi'^\dagger \left(-\frac{\sigma^k}{2m}(F_1(0) + F_2(0))\right)\xi \left(-i\epsilon^{ijk}q^j\tilde{A}_{\text{cl}}^i(\mathbf{q})\right) \\
iM &= -ie2m\xi'^\dagger \left(-\frac{\sigma^k}{2m}(F_1(0) + F_2(0))\right)\xi\tilde{B}^k(\mathbf{q}) \\
iM &= -i\left(-\frac{e}{m}(F_1(0) + F_2(0))2m\xi'^\dagger\frac{\sigma^k}{2}\xi\tilde{B}^k(\mathbf{q})\right)
\end{aligned}$$

where

$$\tilde{B}^k(\mathbf{q}) = \left(-i\epsilon^{ijk}q^j\tilde{A}_{\text{cl}}^i(\mathbf{q})\right)$$

is the Fourier transform of the magnetic field produced by $\mathbf{A}^{\text{cl}}(\mathbf{x})$.

This corresponds to the Born approximation for scattering from a potential

$$\begin{aligned}
V(\mathbf{x}) &= -\frac{e}{m}(F_1(0) + F_2(0))\xi'^\dagger\frac{\sigma^k}{2}\xi B^k(\mathbf{x}) \\
V(\mathbf{x}) &= -\frac{e}{m}(F_1(0) + F_2(0))\xi'^\dagger\frac{\boldsymbol{\sigma}}{2}\xi \cdot \mathbf{B}(\mathbf{x}) \\
V(\mathbf{x}) &= -\langle \boldsymbol{\mu} \rangle \cdot \mathbf{B}(\mathbf{x})
\end{aligned}$$

where

$$\begin{aligned}\langle \boldsymbol{\mu} \rangle &= \frac{e}{m} (F_1(0) + F_2(0)) \xi'^{\dagger} \frac{\boldsymbol{\sigma}}{2} \xi \\ \langle \boldsymbol{\mu} \rangle &= g \frac{e}{2m} \mathbf{S}\end{aligned}$$

where

$$\begin{aligned}g &= 2(F_1(0) + F_2(0)) \\ \mathbf{S} &= \xi'^{\dagger} \frac{\boldsymbol{\sigma}}{2} \xi\end{aligned}$$

The coefficient g is called the Landé g-factor, and since the leading order of perturbation theory gives $F_2(0) = 0$ (and we know that $F_1(0) = 1$ to all orders), we get:

$$g = 2(F_1(0) + F_2(0)) = 2 + 2F_2(0) = 2 + O(\alpha)$$

This is the standard prediction of the Dirac equation. The anomalous magnetic moment is then:

$$a_e = \frac{g - 2}{2} = F_2(0)$$

To calculate that, we need to evaluate the one-loop correction to the vertex function, so we start by deriving the appropriate Green function for the process $\gamma(q) + e^+(p) \rightarrow e^+(p')$:

$$\begin{aligned}|i\rangle &= a_{\mathbf{q}}^{r\dagger} b_{\mathbf{p}}^{t\dagger} |\Omega\rangle \\ |f\rangle &= b_{\mathbf{p}'}^{s\dagger} |\Omega\rangle \\ \langle f|i\rangle &= \langle \Omega | b_{\mathbf{p}'}^s a_{\mathbf{q}}^{r\dagger} b_{\mathbf{p}}^{t\dagger} |\Omega\rangle = \\ &= \langle \Omega | T b_{\mathbf{p}'}^s a_{\mathbf{q}}^{r\dagger} b_{\mathbf{p}}^{t\dagger} |\Omega\rangle = \\ &= \langle \Omega | T \bar{u}^s(\mathbf{p}') \frac{1}{\tilde{S}(p')} \tilde{\psi}(p') \epsilon_{\mu}^{r*}(\mathbf{q}) \frac{q^2}{i} \tilde{A}^{\mu}(-q) \tilde{\bar{\psi}}(-p) \frac{1}{\tilde{S}(-p)} u^t(\mathbf{p}) |\Omega\rangle = \\ &= \bar{u}^s(\mathbf{p}') \frac{1}{\tilde{S}(p')} \epsilon_{\mu}^{r*}(\mathbf{q}) \frac{q^2}{i} \langle \Omega | T \tilde{\psi}(p') \tilde{A}^{\mu}(-q) \tilde{\bar{\psi}}(-p) |\Omega\rangle \frac{1}{\tilde{S}(-p)} u^t(\mathbf{p}) = \\ &= \bar{u}^s(\mathbf{p}') \frac{1}{\tilde{S}(p')} \epsilon_{\mu}^{r*}(\mathbf{q}) \frac{q^2}{i} \tilde{G}(p, p', q) \frac{1}{\tilde{S}(-p)} u^t(\mathbf{p}) =\end{aligned}$$

where:

$$\tilde{G}(p, p', q) = \langle \Omega | T \tilde{\psi}(p') \tilde{A}^{\mu}(-q) \tilde{\bar{\psi}}(-p) |\Omega\rangle$$

is the interacting Green function for the Lagrangian $-\lambda \bar{e} \gamma^{\mu} e A_{\mu}$. In the first order:

$$\begin{aligned}
\tilde{G}(p, p', q) &= \langle \Omega | T \tilde{\psi}(p') \tilde{A}^\mu(-q) \tilde{\bar{\psi}}(-p) | \Omega \rangle = \\
&= \int d^4x \langle 0 | T \tilde{\psi}(p') \tilde{A}^\mu(-q) \tilde{\bar{\psi}}(-p) (-\lambda) \bar{e}(x) \gamma^\rho e(x) A_\rho(x) | 0 \rangle = \\
&= (-\lambda) \int d^4x d\hat{p}' d\hat{q} d\hat{p} e^{i\hat{p}'p' - \hat{q}q - \hat{p}p} \langle 0 | T \psi(\hat{p}') A^\mu(\hat{q}) \bar{\psi}(\hat{p}) \bar{e}(x) \gamma^\rho e(x) A_\rho(x) | 0 \rangle = \\
&= (-\lambda) \int d^4x d\hat{p}' d\hat{q} d\hat{p} e^{i\hat{p}'p' - \hat{q}q - \hat{p}p} D_\rho^\mu(\hat{q} - x) S(\hat{p}' - x) \gamma^\rho S(\hat{p} - x) = \\
&= (-\lambda) (2\pi)^4 \delta(p' - q - p) \tilde{D}_\rho^\mu(q) \tilde{S}(p') \gamma^\rho \tilde{S}(p)
\end{aligned}$$

so the amplitude is:

$$\begin{aligned}
\langle f | i \rangle &= \bar{u}^s(\mathbf{p}') \frac{1}{\tilde{S}(p')} \epsilon_\mu^{r*}(\mathbf{q}) \frac{q^2}{i} (-\lambda) (2\pi)^4 \delta(p' - q - p) \tilde{D}_\rho^\mu(q) \tilde{S}(p') \gamma^\rho \tilde{S}(p) \frac{1}{\tilde{S}(-p)} u^t(\mathbf{p}) = \\
&= (-\lambda) (2\pi)^4 \delta(p' - q - p) \epsilon_\mu^{r*}(\mathbf{q}) u^s(\mathbf{p}') \gamma^\mu u^t(\mathbf{p})
\end{aligned}$$

and we got $\Gamma^\mu = \gamma^\mu$, so $F_1 = 1$ and $F_2 = 0$ in the lowest order. In the next order we get:

$$\begin{aligned}
\tilde{G}(p, p', q) &= (-\lambda) (2\pi)^4 \delta(p' - q - p) \tilde{D}_\rho^\mu(q) \tilde{S}(p') \delta\Gamma^\rho \tilde{S}(p) \\
\delta\Gamma^\mu &= \int \frac{d^4k}{(2\pi)^4} \tilde{D}_{\nu\rho}(k - p) (-ie\gamma^\nu) \tilde{S}(k') \gamma^\mu \tilde{S}(k) (-ie\gamma^\rho)
\end{aligned}$$

Now we can write:

$$\begin{aligned}
&\bar{u}(p') \Gamma^\mu(p', p) u(p) = \bar{u}(p') (\gamma^\mu + \delta\Gamma^\mu) u(p) \\
&\bar{u}(p') \delta\Gamma^\mu(p', p) u(p) = \int \frac{d^4k}{(2\pi)^4} \tilde{D}_{\nu\rho}(k - p) \bar{u}(p') (-ie\gamma^\nu) \tilde{S}(k') \gamma^\mu \tilde{S}(k) (-ie\gamma^\rho) u(p) = \\
&= \int \frac{d^4k}{(2\pi)^4} \frac{-ig_{\nu\rho}}{(k - p)^2 + i\epsilon} \bar{u}(p') (-ie\gamma^\nu) \frac{i(\not{k}' + m)}{k'^2 - m^2 + i\epsilon} \gamma^\mu \frac{i(\not{k} + m)}{k^2 - m^2 + i\epsilon} (-ie\gamma^\rho) u(p) = \\
&= 2ie^2 \int \frac{d^4k}{(2\pi)^4} \frac{\bar{u}(p') (\not{k} \gamma^\mu u \not{k}' + m^2 \gamma^\mu - 2m(k + k')^\mu) u(p)}{((k - p)^2 + i\epsilon)(k'^2 - m^2 + i\epsilon)(k^2 - m^2 + i\epsilon)} = \dots = \\
&= 2ie^2 \int \frac{d^4l}{(2\pi)^4} \int_0^1 dx dy dz \delta(x + y + z - 1) \frac{2}{D^3} \bar{u}(p') \left(\gamma^\mu (-\frac{1}{2}l^2 + (1 - x)(1 - y)q^2 + (1 - 4z + z^2)m^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m} (2m^2z(1 - z)) \right) u(p) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x + y + z - 1) \bar{u}(p') \left(\gamma^\mu \left[\log \frac{z\Lambda^2}{\Delta} + \frac{1}{\Delta} ((1 - x)(1 - y)q^2 + (1 - 4z + z^2)m^2) \right] + \frac{i\sigma^{\mu\nu}q_\nu}{2m} \left[\frac{1}{\Delta} 2m^2z(1 - z) \right] \right) u(p)
\end{aligned}$$

where

$$\begin{aligned}
k' &= k + q \\
D &= l^2 - \Delta + i\epsilon \\
\Delta &= -xyq^2 + (1 - z)^2m^2 > 0
\end{aligned}$$

So the expressions for the form factors are:

$$\begin{aligned}
F_1(q^2) &= 1 + \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\log \frac{z\Lambda^2}{\Delta} + \frac{1}{\Delta} ((1-x)(1-y)q^2 + (1-4z+z^2)m^2) \right] + O(\alpha^2) \\
F_2(q^2) &= \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{1}{\Delta} 2m^2 z(1-z) \right] + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2m^2 z(1-z)}{m^2(1-z)^2 - q^2 xy} \right] + O(\alpha^2)
\end{aligned}$$

F_1 contains both ultraviolet and infrared divergencies. To cure the infrared divergence, we add a term $\mu^2 z$ to Δ . To cure the ultraviolet divergence, we make the substitution:

$$F_1(q^2) \rightarrow F_1(q^2) - \delta F_1(0)$$

where δF_1 is the first order (in α) correction to F_1 (i.e. $F_1 = 1 + \delta F_1 + O(\alpha^2)$):

$$\delta F_1(0) = \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\log \frac{z\Lambda^2}{\Delta(q^2=0)} + \frac{1}{\Delta(q^2=0)} (1-4z+z^2)m^2 \right]$$

so the corrected F_1 is:

$$\begin{aligned}
F_1(q^2) &= 1 + \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\log \frac{z\Lambda^2}{\Delta} + \frac{1}{\Delta} ((1-x)(1-y)q^2 + (1-4z+z^2)m^2) + \right. \\
&\quad \left. - \log \frac{z\Lambda^2}{\Delta(q^2=0)} - \frac{1}{\Delta(q^2=0)} (1-4z+z^2)m^2 \right] + O(\alpha^2) = \\
&= 1 + \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\log \frac{m^2(1-z)^2}{m^2(1-z)^2 - q^2 xy} + \left(\frac{(1-x)(1-y)q^2 + (1-4z+z^2)m^2}{m^2(1-z)^2 - q^2 xy + \mu^2 z} \right) + \right. \\
&\quad \left. - \frac{(1-4z+z^2)m^2}{m^2(1-z)^2 + \mu^2 z} \right] + O(\alpha^2)
\end{aligned}$$

Neither the ultraviolet nor the infrared divergence affects $F_2(q^2)$, so we just set $q = 0$:

$$\begin{aligned}
F_2(0) &= \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2m^2 z(1-z)}{m^2(1-z)^2} \right] + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dx dy dz \delta(x+y+z-1) \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dy \int_0^1 dz \theta(1-(1-y-z)) \theta((1-y-z)-0) \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dy \int_0^1 dz \theta(y+z) \theta(1-y-z) \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dy \int_0^1 dz \theta(1-y-z) \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dz \int_0^{1-z} dy \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dz (1-z) \frac{2z}{1-z} + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} \int_0^1 dz 2z + O(\alpha^2) = \\
&= \frac{\alpha}{2\pi} + O(\alpha^2)
\end{aligned}$$

Thus we get the correction to the g -factor of the electron:

$$a_e = \frac{g-2}{2} = F_2(0) = \frac{\alpha}{2\pi} \approx 0.0011614$$

Code:

```
>>> from math import pi
>>> alpha = 1/137.035999049
>>> a_e = alpha / (2*pi)
>>> a_e
0.0011614097331824923
```

Experiments give $a_e = 0.00115965218073 \pm 0.000000000000028$ ([arXiv:1412.8284](#), eq. (1)).

Higher order corrections from QED can also be calculated:

$$a_e = A_1 \left(\frac{\alpha}{\pi} \right) + A_2 \left(\frac{\alpha}{\pi} \right)^2 + A_3 \left(\frac{\alpha}{\pi} \right)^3 + A_4 \left(\frac{\alpha}{\pi} \right)^4 + \dots$$

we already know that $A_1 = \frac{1}{2}$. See for example [hep-ph/9410248](#) for the expression for A_2 :

$$\begin{aligned}
A_2 &= \frac{197}{144} + \frac{3}{4} \zeta(3) - \frac{1}{2} \pi^2 \log(2) + \frac{1}{12} \pi^2 = \\
&= -0.328478965579 \dots
\end{aligned}$$

Code:

```
>>> from sympy import zeta, S, log
>>> A_2 = S(197)/144 + zeta(2)/2 + 3*zeta(3)/4 - 3*zeta(2) * log(2)
```

```
>>> A_2.n()
-0.328478965579194
```

See [hep-ph/9602417](https://arxiv.org/abs/hep-ph/9602417) for the A_3 term:

$$A_3 = \frac{28259}{5184} - \frac{215}{24}\zeta(5) + \frac{100}{3} \left(\sum_{n=1}^{\infty} \frac{1}{2^n n^4} - \frac{1}{24}\pi^2 \log^2(2) + \frac{1}{24}\log^4(2) \right) + \\ + \frac{139}{18}\zeta(3) - \frac{298}{9}\pi^2 \log(2) + \frac{83}{72}\pi^2 \zeta(3) + \frac{17101}{810}\pi^2 - \frac{239}{2160}\pi^4 = \\ = 1.181241456\dots$$

Code:

```
>>> from sympy import pi, zeta, S, log, sum, var, oo
>>> var("n")
n
>>> a4 = sum(1/(2**n * n**4), (n, 1, oo))
>>> A_3 = 83*pi**2*zeta(3)/72 - 215*zeta(5)/24 + 100*(a4 + log(2)**4/24 - \
...      pi**2*log(2)**2/24)/3 - \
...      239*pi**4/2160 + 139*zeta(3)/18 - 298 * pi**2 * log(2)/9 + \
...      17101 * pi**2 / 810 + S(28259)/5184
>>> A_3.n()
1.18124145658720
```

Higher terms are only known numerically. The A_4 and A_5 terms can be found in [arXiv:1412.8284](https://arxiv.org/abs/1412.8284):

$$A_4 = -1.91298(84) \\ A_5 = 7.795(336)$$

We can now sum a_e up to a given order by the following script:

```
from sympy import pi, zeta, S, log, summation, var, oo
var("n")
a4 = summation(1/(2**n * n**4), (n, 1, oo))
A1 = S(1)/2
A2 = S(197)/144 + zeta(2)/2 + 3*zeta(3)/4 - 3*zeta(2) * log(2)
A3 = 83*pi**2*zeta(3)/72 - 215*zeta(5)/24 + 100*(a4 + log(2)**4/24 - \
      pi**2*log(2)**2/24)/3 - \
      239*pi**4/2160 + 139*zeta(3)/18 - 298 * pi**2 * log(2)/9 + \
      17101 * pi**2 / 810 + S(28259)/5184
A4 = -1.91298
A5 = 7.795
alpha = 1/137.035999049
a_e_exp = 0.00115965218073
a_e_exp_err = 0.00000000000028
a_e_other = 0.000000000000448
A = [A1, A2, A3, A4, A5]
a_e = []
for i in range(len(A)):
    a_e.append((A[i]*(alpha/pi)**(i+1)).n())
print "===== "
print "Order      :math:`a_e`"
print "===== "
for i in range(len(A)):
    print "%d          %16.14f" % (i+1, sum(a_e[:i+1]))
print "Other          %16.14f" % a_e_other
```

```

print "Total      %16.14f" % (sum(a_e) + a_e_other)
print "Experiment %16.14f" % a_e_exp
print "Difference %16.14f" % abs(sum(a_e) + a_e_other - a_e_exp)
print "Exp. err   %16.14f" % a_e_exp_err
print "===== "

```

and obtain the following table:

Order	a_e
1	0.00116140973318
2	0.00115963742812
3	0.00115965223232
4	0.00115965217663
5	0.00115965217716
Other	0.00000000000448
Total	0.00115965218164
Experiment	0.00115965218073
Difference	0.00000000000091
Exp. err	0.00000000000028

The “Other” line are contributions from the dependence on the muon and tau particle masses, the hadronic vacuum-polarization, the hadronic light-by-light-scattering and the electroweak contribution (see [arXiv:1412.8284](https://arxiv.org/abs/1412.8284)). The “Difference” line is the difference from the theory (the “Total” line) and experiment. The “Exp. err” line is the experimental error.

At this level of accuracy, the uncertainty of the exact value of α is the primary cause of the difference from experiment, and one can use this result to predict a more accurate value for α , assuming that QED and the standard model are valid.